User Guide
Self-Service Upgrade Tool v1.3

Professional Services Organization

Palo Alto Networks, Inc.

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Introduction

Purpose

The primary purpose of this User Guide is to provide Prisma Cloud users with comprehensive, step-by-step instructions and essential information required to successfully utilize the Self-Service Upgrade Tool as part of their upgrade to Cortex Cloud.

Specifically, this guide aims to:

- **Outline Prerequisites:** Detail necessary permissions, access credentials (for both Prisma Cloud and Cortex Cloud), network requirements, and any preparation needed.
- **Guide Installation/Access:** Provide instructions on how to install and access the tool.
- **Configure Upgrade Workflows:** Clearly outline how to configure the tool, including connecting to source (Prisma Cloud) and destination (Cortex Cloud) environments, and selecting upgrade options or scope.
- **Instruct on Execution:** Offer step-by-step guidance on initiating, monitoring, and managing the upgrade process via the tool interface.
- **Validate Outcomes:** Explain how users can verify the successful completion and accuracy of the upgrade.
- **Aid Troubleshooting:** Provide information on common issues, error messages, log interpretation, and basic troubleshooting steps users can take.
- **Define Scope and Limitations:** Clarify exactly what data, configurations, or settings the tool is designed to migrate and any known limitations.

Ultimately, this guide serves to empower users to effectively operate the tool independently, ensuring an accurate upgrade from Prisma Cloud to Cortex Cloud.

Executive Summary

The Self-Service Upgrade Tool provides an automated, user-driven solution to streamline the upgrade from Prisma Cloud to Cortex Cloud. This tool empowers security and IT teams to efficiently migrate relevant configurations and settings, minimizing manual effort and accelerating the adoption of Cortex Cloud.

Key Benefits

- **Accelerated upgrade:** Significantly reduces the time required for upgrade compared to manual processes, enabling faster onboarding onto Cortex Cloud.
- **Reduced Cost and Effort:** Automates repetitive tasks, freeing up valuable technical resources and minimizing potential consulting costs associated with manual upgrade projects.
- **Minimized Risk and Increased Accuracy:** Reduces the likelihood of human error during configuration transfer, ensuring critical settings and configurations are migrated correctly, thereby maintaining security visibility continuity.

Relationship to Cortex Upgrade Helper

It is important to distinguish this Self-Service Upgrade Tool from the Upgrade Helper functionality built into the Cortex Cloud platform. While the native Upgrade Helper typically facilitates internal platform upgrades, component upgrades within the Cortex ecosystem, or specific user onboarding workflows, this dedicated tool addresses the distinct scenario of upgrading relevant settings and configurations from Prisma Cloud, specifically into Cortex Cloud. Therefore, this tool serves as a complementary capability, purpose-built to handle the unique requirements of upgrading from Prisma Cloud, rather than replacing the functions provided by the standard Cortex Upgrade Helper for platform management.

Prerequisites

Before utilizing the Self-Service Upgrade Tool, ensuring all prerequisites are met is crucial for an accurate upgrade process. This preparation spans the source (**Prisma Cloud**), the destination (**Cortex Cloud**), and the features being migrated.

Prisma Cloud Environment

- **Active Subscription:** A valid Prisma Cloud Enterprise subscription is required.
- **API Access:** You must generate a **Prisma Cloud Access Key**, consisting of an Access Key ID and a Secret Key. The recommendation is to create a service account for the process.
- **Permissions:** The Access Key must be associated with the **System Admin** Prisma Cloud role that has sufficient **permissions** to access the specific settings, configurations, or other data points targeted for upgrade.
- **Connectivity:** Network access must be available from the environment where the upgrade tool operates to the relevant **Prisma Cloud API** endpoint and the Prisma Cloud Compute Console URL.

Cortex Cloud Environment

- **Active Tenant:** A valid and provisioned **Cortex Cloud** tenant is necessary with the Application Security features.
- **API Access Credentials:** You need appropriate **Cortex Cloud API access credentials**. Depending on the specific Cortex service being targeted by the upgrade tool, this may involve generating a specific type of API Key, Key ID, or service account token through the Cortex Cloud platform interface.
- **Permissions:** The created Cortex Cloud credentials must be with the role **Instance Administrator** that has sufficient **permissions** to access the specific settings, configurations, or other data points targeted for upgrade. Refer to the **User Roles and Groups** for details.
- **API Endpoint:** The correct **API endpoint URL** for your specific Cortex Cloud tenant.
- **Connectivity:** Network access must be permitted from the upgrade tool's environment to the designated **Cortex Cloud API endpoints**.

Resource Pre-onboarding, Preparation and Considerations

- **Define upgrade Scope:** Clearly identify and document the specific application security configurations within Prisma Cloud that are intended for upgrade.

- **Verify Source Data:** Confirm that the resources identified in the scope are present, accurate, and properly configured within Prisma Cloud before initiating the upgrade.
- **VCS Integrations:** Ensure that you have already onboarded your VCS into Cortex Cloud, and that repositories are already configured and showing active on the tool. This is a crucial step to ensure the correct functionality of the tool.

Considerations

Cloud Security Posture Management

- **Resource lists:**
  - Compute Asset Groups defined in Prisma's Resource lists section operate under the same logic and principles as Runtime Collections. Since there is no mapping of policy usage for these items to determine use cases, the equivalent combinations permitted for collections, which depend on usage, will be applied in this scenario.
  - Resource lists based on tags are designed to match any resource that has at least one of the specified tag values (a Logical OR operation). The matching process uses a wildcard comparison, which is the equivalent of a "contains" operation.
- **Account Groups:**
  - For account groups defined as a parent, the resulting asset group's logic will encompass not only the directly linked accounts but also any defined child accounts. All these accounts will be combined using a logical OR within a single asset group.
  - Assets will be included in the group if they belong to at least one of the account parts within the result groups. Successful inclusion requires the asset cloud account ID to contain the defined values on Prisma.

Cloud Workload Protection

- **Collections:**
  - Only collections assigned to supported policies will be copied as asset groups into Cortex Cloud. Collections that are not currently in use are outside the scope of this execution.
  - For each policy type in Prisma Cloud where a collection is assigned, a corresponding asset group will be generated in Cortex. Therefore, if a single collection is assigned to three distinct policies, three asset groups will be created in Cortex.
  - If a wildcard is included in the collection, the asset group's query will execute a "contains" operation.

  - The application of conditions to each asset group is determined by the policy it's used within. It's important to note that even when asset groups are created from the same collection, their parameters might reference distinct resources.
  - Apps IDs will not be supported
- **Vulnerability Policies:**
  - In the base execution of the feature, one policy will be copied as 2 in Cortex, one for alert behaviour and the second one for block behaviour.
  - Blocking policies defined in Policies are going to be copied only in alert behavior into Cortex.
  - To ensure policies are copied into Cortex, the asset groups must be an exact match. If an asset group is absent according to the Prisma original definition, the policy will not be generated. It is recommended to use the Collections feature beforehand to create asset groups and confirm their existence before reviewing the current feature.
  - If this feature is selected along with the Collections feature. The collections module will complete its whole flow before the transformation of these policies start.
- **Compliance Policies:**
  - Cortex currently does not support compliance checks for images\containers.
  - Each Prisma compliance rule supported by the tool that contains one of the malware rules will be copied to a new Cortex Malware policy.
  - To ensure policies are copied into Cortex, the asset groups must be an exact match. If an asset group is absent according to the Prisma original definition, the policy will not be generated. It is recommended to use the Collections feature beforehand to create asset groups and confirm their existence before reviewing the current feature.
  - If this feature is selected along with the Collections feature. The collections module will complete its whole flow before the transformation of these policies start.
- **Registry Scan Inventory:**
  - This feature is not supported in Cortex. Its purpose in the tool is to generate an output file containing existing Prisma configurations for mapping purposes.
  - The result file can be found in the directory specified in the “Outputs” section of this document.

Application Security

- **Validate Secrets and Secrets History**
  - Validate Secrets will configure the scan configuration of each one of the onboarded repos on Cortex Cloud. This update will be performed on *all* the repositories onboarded, overriding any manual configuration established before executing the tool.
  - For improved behaviour, execute this feature after a successful migration of the repositories.
- **Enforcement Rules**
  - Only default Enforcement rules configurations are part of the scope of the feature. Exceptions will not be migrated to Cortex Cloud, but post execution modification can be performed on the tenant to adjust the exceptions.
  - Hardfail enforcement rules will be migrated with alert behaviour mode to avoid operational disrupts.
  - Developer Suppressions will be automatically configured on the generated policies, based on the current status of this feature in Prisma Cloud.
- **Repository Selection**
  - Only the integrations marked with the “Manual Selection” feature on Prisma Cloud are going to be considered as part of the scope.
  - The destination integrations must be onboarded manually before executing the tool. If there is no corresponding integration on Cortex Cloud, once the tool is executed no upgrade will be performed to the integration even if it was within the scope of the feature
  - Integrations part of the scope will override any previous configuration existing on Cortex Cloud to assume the same settings as Prisma Cloud. All updated integrations will have Manual Selection and the same set of repositories as defined on Prisma Cloud.
  - For GitHub and Bitbucket integration previously, Prisma Cloud permitted only one integration. Cortex Cloud enhances this by allowing multiple integrations simultaneously. Because of this change, ensure all target repositories are correctly selected. The system will select specified repositories if their names or unique identifiers are found within Cortex Cloud integration and if the integration itself can be matched with the ones in Prisma Cloud, otherwise, they will be omitted from the upgrade.
  - If there's more than one integration with the same identifier in Cortex Cloud that matches an integration within the scope of the tool, the tool will update each one of them. However, only the last one updated will remain with the repositories onboarded.
- **OOTB Policies Labels**

- Note that labels on out-of-the-box (OOTB) policies are overwritten during the upgrade process. If an OOTB policy has previously defined custom labels, the upgrade tool will overwrite them.
- Only Build policies are part of the scope of this upgrade.

Meeting these prerequisites and considerations beforehand will significantly streamline the upgrade process facilitated by the Self-Service Upgrade Tool into your **Cortex Cloud** environment.

Setup

This section outlines the essential preparatory steps before you begin configuring the Self-Service Upgrade Tool. Proper setup is vital for the tool to operate correctly and perform the upgrade successfully.

It is critical, before launching the tool or proceeding to the next step, to ensure you have thoroughly reviewed and fulfilled all requirements detailed in the 'Prerequisites' section of this guide. This verification step confirms you have:

- Active subscriptions for both platforms.
- Necessary network connectivity.
- Obtained the specific API Credentials (Access Keys, API Keys, Secret Keys, IDs, FQDNs/URLs) and possess the required user permissions within both your Prisma Cloud and Cortex Cloud environments.
- Properly installed the Self-Service Upgrade Tool executable and dependencies, depending on the OS (Windows or macOS).

Completing these prerequisites beforehand is mandatory for the tool to authenticate and execute the upgrade tasks.

Installation

To obtain the Self-Service Upgrade Tool, navigate to the designated download location at LIVEcommunity.

The download consists of a single ZIP file. This archive contains the necessary executable, providing a version compatible with all supported Operating Systems (e.g., Windows, macOS). Once downloaded, you will need to extract the contents of this ZIP file to a local directory on your system before running the appropriate executable for your OS.

Running the Installation Scripts

Inside the ZIP file you downloaded, you will find the main executable, tailored for the specific Operating System you have selected:

- **Windows:** self-upgrade-tool.exe
- **macOS:** self-upgrade-tool.app (There are two for macOs depending on the architecture)

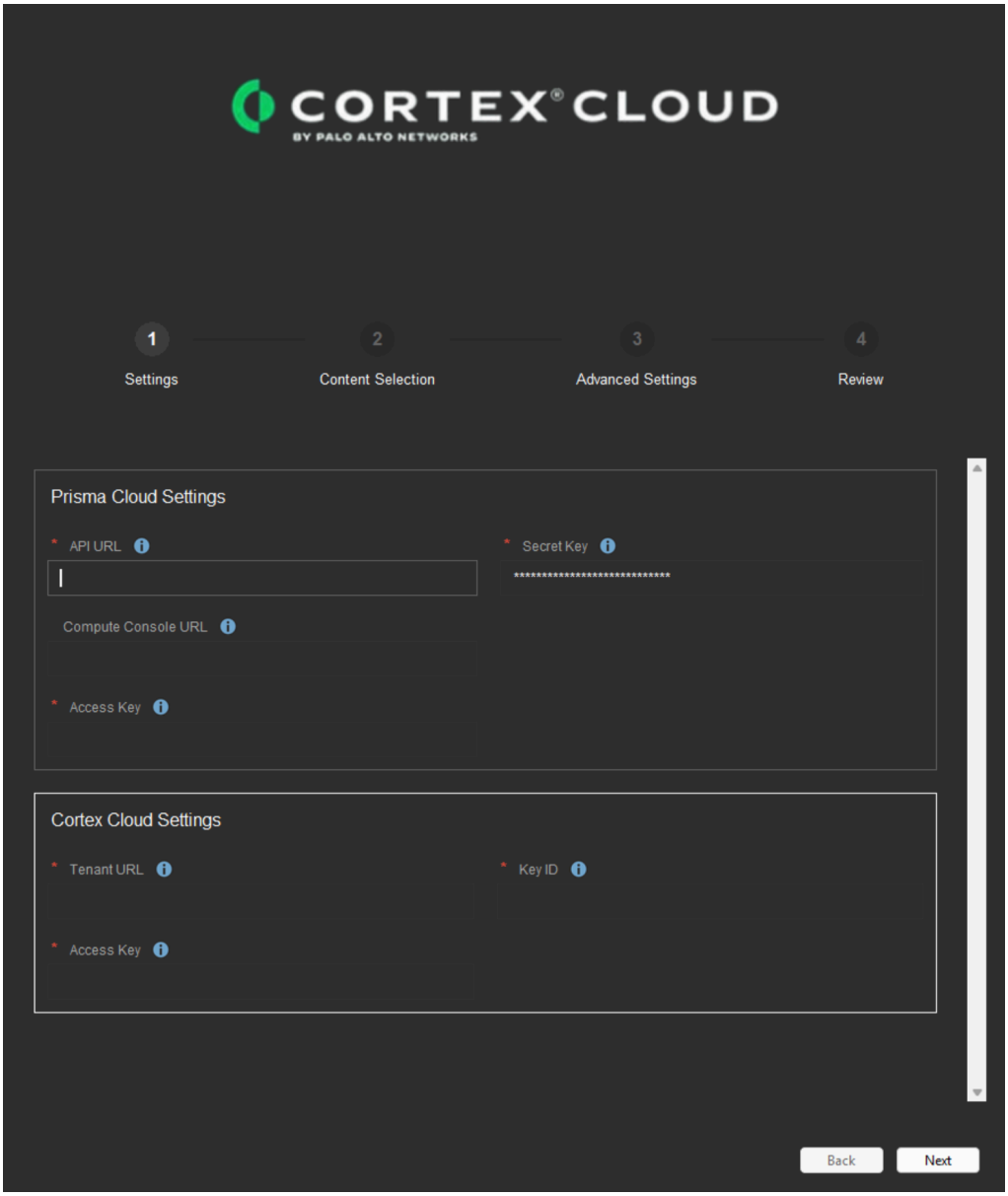
Note: Ensure that you have enough permissions to run the application.

This executable will launch the **Self-Service Upgrade Tool**. Follow any on-screen prompts or refer to the tool's interface to proceed with the upgrade process.

Settings

Once you launch the Self-Service Upgrade Tool, the initial screen presented by the wizard is the **Settings** tab. In this tab provide the fundamental connection details that allow the tool to communicate securely with both your source **(Prisma Cloud)** and destination **(Cortex Cloud)** platforms. These details should correspond to the information gathered while fulfilling the prerequisites.

This view begins with a "Requirements" section that outlines the necessary prerequisites and important considerations for using the tool. This section also summarizes the tool's scope and expected outputs. The information presented is obtained directly from this document.



Prisma Cloud

In the **Prisma Cloud Settings** section of the **Settings** screen of the upgrade tool, you must also provide the connection details for your Prisma Cloud environment. Tooltips are available for each field within the tool interface, or by hovering over the information icon to provide specific guidance on obtaining or formatting the required values.

Follow these instructions to retrieve the required Prisma Cloud information:

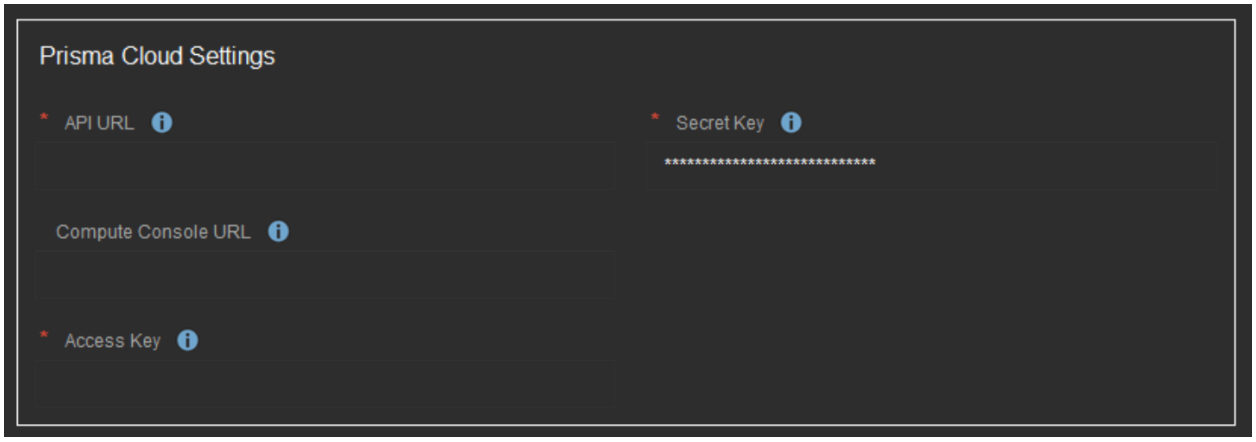
1. **Prisma Cloud Access Key ID and Secret Key**
    - **Description**: These credentials authenticate the upgrade tool's API access to your Prisma Cloud account. They are generated together as a pair.
    - **Generation Process**:
        i. Log in to your Prisma Cloud platform.
        ii. Navigate to: **Settings > Access Control > Access Keys**.
        iii. Click **Add > Access Key**.
        iv. Follow the prompts to configure the key name and assign an appropriate role with sufficient permissions (refer to the Prerequisites section for details).
        v. **Critical**: Upon successful creation, Prisma Cloud will display the **Access Key ID and the Secret Key only once**. **You must copy both values immediately and store them securely**, especially the Secret Key which cannot be viewed again.
        vi. Paste the copied **Access Key ID** into the **Access Key** field and the copied **Secret Key** into the **Secret Key** field in the upgrade tool.

2. **Prisma Cloud API URL**
    - **Description**: The base URL for the Prisma Cloud API endpoint specific to your tenant's deployment region or instance.
    - **Generation Process**: You must identify the correct API endpoint URL for your specific Prisma Cloud instance. Consult the official Prisma Cloud API documentation to determine the accurate URL designated for API access.
    - Paste the correct **API URL** into the **API URL** field in the upgrade tool.
    - Note: If you encounter an authentication error in the Prisma tool when your API URL is https://app.prismacloud.io, make sure to use https://app0.prismacloud.io to resolve the issue.

3. **Compute Console URL**
    - **Description**: This URL will be used to gather the information and configurations from the Cloud Workload Protection module in Prisma Cloud.

i. Log in to your Prisma Cloud platform.
ii. Select the Run Time Module.
iii. Navigate to **Manage > System > Utilities > Path to Console**
iv. Copy the value available in this section.
v. Past the value in the “Compute Console URL” field in the upgrade tool.

Verify the accuracy of all entered credentials before proceeding.



Cortex Cloud

Within the upgrade tool's **Cortex Cloud Settings** screen, you must provide the necessary credentials to establish a connection with your Cortex Cloud environment. Tooltips are available for each field within the tool interface, or by hovering over the **?** icon to provide specific guidance on obtaining or formatting the required values.

Follow these steps to retrieve the necessary information from Cortex Cloud:

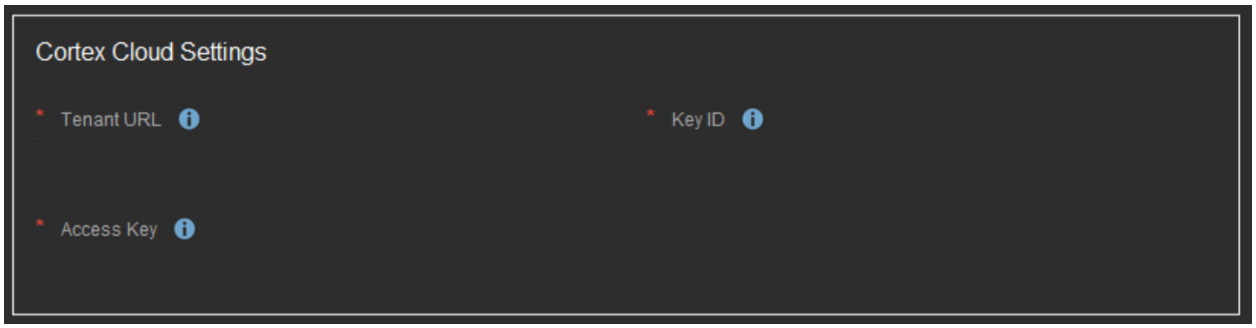
1. **Cortex Cloud API URL/FQDN:**
    - **Description**: This is the specific API endpoint URL required for programmatic communication with your Cortex Cloud instance.
    - **Generation Process**: This URL or FQDN (Fully Qualified Domain Name) is typically provided by Cortex Cloud during or after the API Key generation process (see step 2 below), often within the key details or API usage examples. Copy the exact URL/FQDN provided by Cortex Cloud, ensuring it matches the format specified in the tool's tooltip

(e.g., https://api-<your_unique_fqdn>.paloaltonetworks.com). Paste this value into the corresponding field in the upgrade tool.

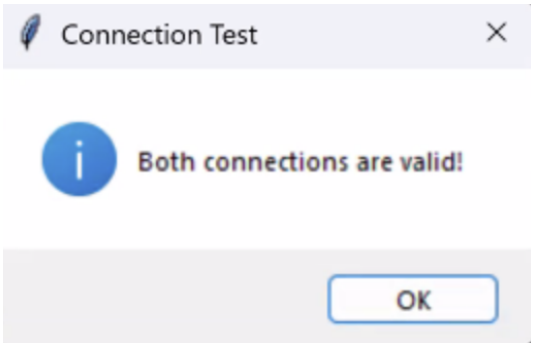
2. **Cortex Cloud API Key:**
    - **Description**: This credential authenticates the upgrade tool's API requests to Cortex Cloud.
    - **Generation Process**:
        i. Navigate within Cortex Cloud to: **Settings > Configurations > Integrations > API Keys**.
        ii. Select **+ New Key**.
        iii. Configure the key settings as required, ensuring it is assigned a role with sufficient permissions (refer to the Prerequisites section for details). Note: Make sure the security level is set to “Standard”.
        iv. **Critical**: The generated API Key value is displayed only once immediately after creation. **Copy this key value immediately and store it securely.** It cannot be retrieved later.
        v. Paste the copied API Key into the designated field in the upgrade tool.

3. **Cortex Cloud Key ID**
    - **Description**: This value uniquely identifies the specific API Key being used.
    - **Generation Process**: After generating the key, return to the list view at **Settings > Configurations > Integrations > API Keys** in Cortex Cloud. Locate the newly created key and copy its corresponding ID (usually displayed as a numerical value in the table). Paste this **ID** into the **'Key ID'** field in the **upgrade** tool.

Verify the accuracy of all entered credentials before proceeding.



After setting up the tool, click on the **Next** button on the bottom right corner of the wizard and the credentials will be tested and validated.



If both connections are successful you will see the green checkmark on the bottom of each of the sections showing that the connections are established and ready to run the upgrade.

If any of the entered values are not valid or improperly entered, the tool will prompt with an error pointing to the settings that failed.

Running Upgrade

Content Selection

Upon successfully completing the configuration in the **Settings** tab and clicking **Next**, you will proceed to the **Content Selection** section. This critical step allows you to precisely choose which Prisma Cloud configuration elements you wish to copy into your Cortex Cloud environment.

For each available feature listed below, you can access helpful details via a tooltip. Simply hover your mouse cursor over the information icon beside it. The tooltip will explain the scope of that specific feature (detailing what settings are included) and the expected outcome (how it will appear or function within Cortex Cloud after upgrade).

Review the information in the tooltips carefully to understand the impact of migrating each item. Once you have decided, select the checkbox next to each feature you want to include in this upgrade process:

Cloud Security Posture Management

- **Account Groups:** Copy Prisma Cloud Account Groups to Cortex Cloud as asset groups. Parent Groups will include in the result asset groups all the child accounts.
- **Resource Lists:** Copy Prisma Cloud Resource Lists to Cortex Cloud as asset groups. Supported types of resource lists include Tag definition, Azure Resource Groups, and Compute Access Groups.

Runtime Security

- **Collections:** Copy Prisma Cloud collections to Cortex Cloud as asset groups. This feature is limited to collections currently assigned to compliance or vulnerability policies in Prisma. An asset group will be generated for each policy it is applied to.
- **Vulnerability Policies:** Copy Prisma Cloud vulnerability policies to Cortex Cloud as Vulnerability Management Policies. This process will create two policies: one for alert behavior and another for blocking behavior if the corresponding thresholds were configured in Prisma Cloud. The original priority order of the policies will be preserved in Cortex after the transfer.

  Note: If existing policies were present in Cortex prior to this operation, the copied policies will be appended in the same order after the last existing policy. CVE exclusions are supported by default.

- **Compliance Policies:** Copy Prisma Cloud compliance policies to Cortex Cloud as Cloud Workload Policies. This action generates two policies: one for alert behavior and one for block behavior. If any selected Prisma rules in the original policy were classified as a malware rule, a corresponding malware policy will also be created based on the same principle.

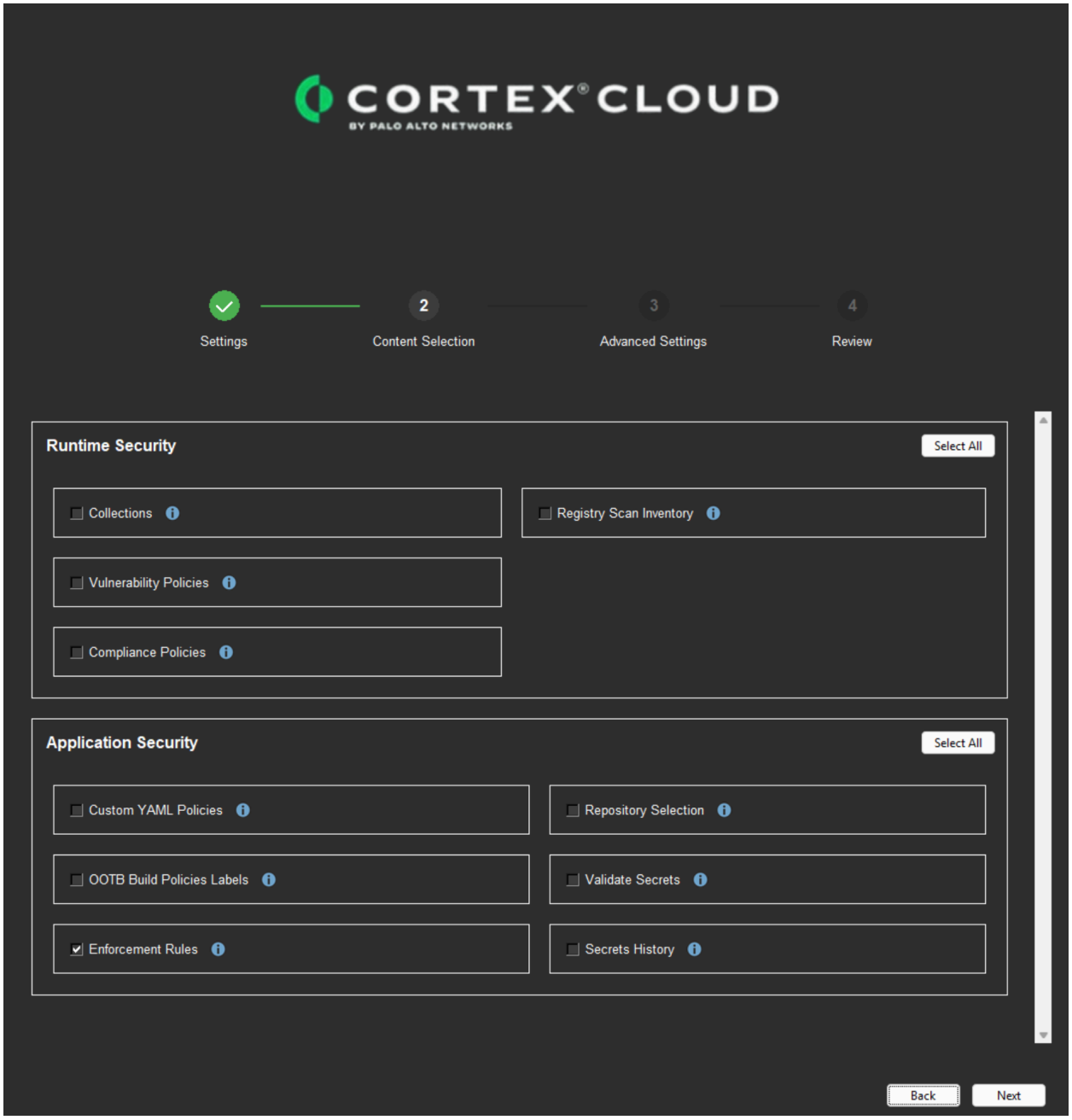
  Note: In this release, the update endpoint presented some issues. Consequently, while the first execution of this feature will succeed, subsequent executions may result in warnings or errors.

- **Registry Scan Inventory:** Generates an output file containing key configurations and elements from the Registry Scan Settings in JSON format. This item is saved to the same directory as other execution outputs.

Application Security

- **Custom YAML Policies:** Copy custom policy definitions written in YAML, these policies include the ones created via Visual editor and Code editor.
- **Repository Selection:** If an integration uses manual repository selection in Prisma Cloud, the process ensures the same repositories are selected for the corresponding integration in Cortex Cloud.
- **Enforcement Rules:** Copy the actual enforcement rules on Prisma Cloud and create the policies on Cortex Cloud.
  Note: All the Hard-Fail definitions on Prisma Cloud will be copied into Cortex Cloud in alerting mode to avoid any disruptions on the environments.
- **OOTB Build Policies Labels:** Copy the labels applied to out-of-the-box build policies into Cortex Cloud.
- **Validate Secrets:** If the feature is enabled in Prisma Cloud this will copy the settings into Cortex Cloud.
- **Secrets History:** If the feature is enabled in Prisma Cloud this will copy the settings into Cortex Cloud.

Ensure your selections accurately reflect the configurations and data you need in Cortex Cloud before proceeding to the next step.



Advanced View

Based on the selections made in the previous step, this view will show additional adjustments to certain functionalities, allowing for the scope and behavior of these features to be adjusted during the current execution. If none of the selected options

have associated advanced options, the tool will display a message indicating this situation and you can proceed to the next step.

Cloud Security Posture Management

- **Account Groups:**
  - Migrate all Account Groups: Set the scope of the feature to all the identified account groups.
  - Select specific names: Choose the name of the Account groups you wish to copy into Cortex.
- **Resource Lists:**
  - Migrate all Resource Lists: Set the scope of the feature to all the identified resource lists.
  - Select specific names: Choose the name of the resource lists you wish to copy into Cortex.

Cloud Workload Protection

- **Compliance Policies**
  - Generate One Policy By Severity: Independent policies will be generated grouping rules by severity. This allows issues to be generated with their original assigned severity while the "Assume Severity of the Rule" capability is enabled in Cortex. This feature can create up to four new policies from a single existing one.
- **Vulnerability Policies**
  - Include Tags in Vulnerabilities: Adjust the scope of the rules or include additional resources to consider the exclusions defined in Prisma Tags. Cortex Cloud lacks a direct equivalent for tags; therefore, the tool will generate multiple asset groups and supplementary rules to replicate these exclusions. Warning: activating this feature might substantially increase the number of policies and asset groups generated.

Application Security

- **Custom YAML Policies**
  - Migrate all Policies: Set the scope of the feature to all the identified policies.
  - Choose labels from list: Choose the labels you wish to copy into Cortex. Only policies in Prisma that are tagged with the selected label will be included in the execution.

    - Choose names from list: Choose the name of the policies you wish to copy into Cortex.
    - Upgrade Only Enable Policies: Selecting this option will copy only the active custom policies from Prisma Cloud to Cortex Cloud.
- **Enforcement Rules**
    - Group Policies: Selection this option will group the policies that have the same actions, conditions and triggers in a single policy. The policy's name will reflect the elements that were combined into a single policy.
    - Configure issues in created Policies: Enabling this option will configure the policy to create issues based on the established conditions.
- **OOTB Build Policies Labels**
    - Migrate all Policies: Set the scope of the feature to all the identified policies.
    - Choose labels from list: Choose the labels you wish to copy into Cortex. Only policies in Prisma that are tagged with the selected label will be included in the execution.
    - Choose names from list: Choose the name of the policies you wish to copy into Cortex.
    - Upgrade Only Enable Policies: Selecting this option will copy only the active OOTB policies from Prisma Cloud to Cortex Cloud.



Review

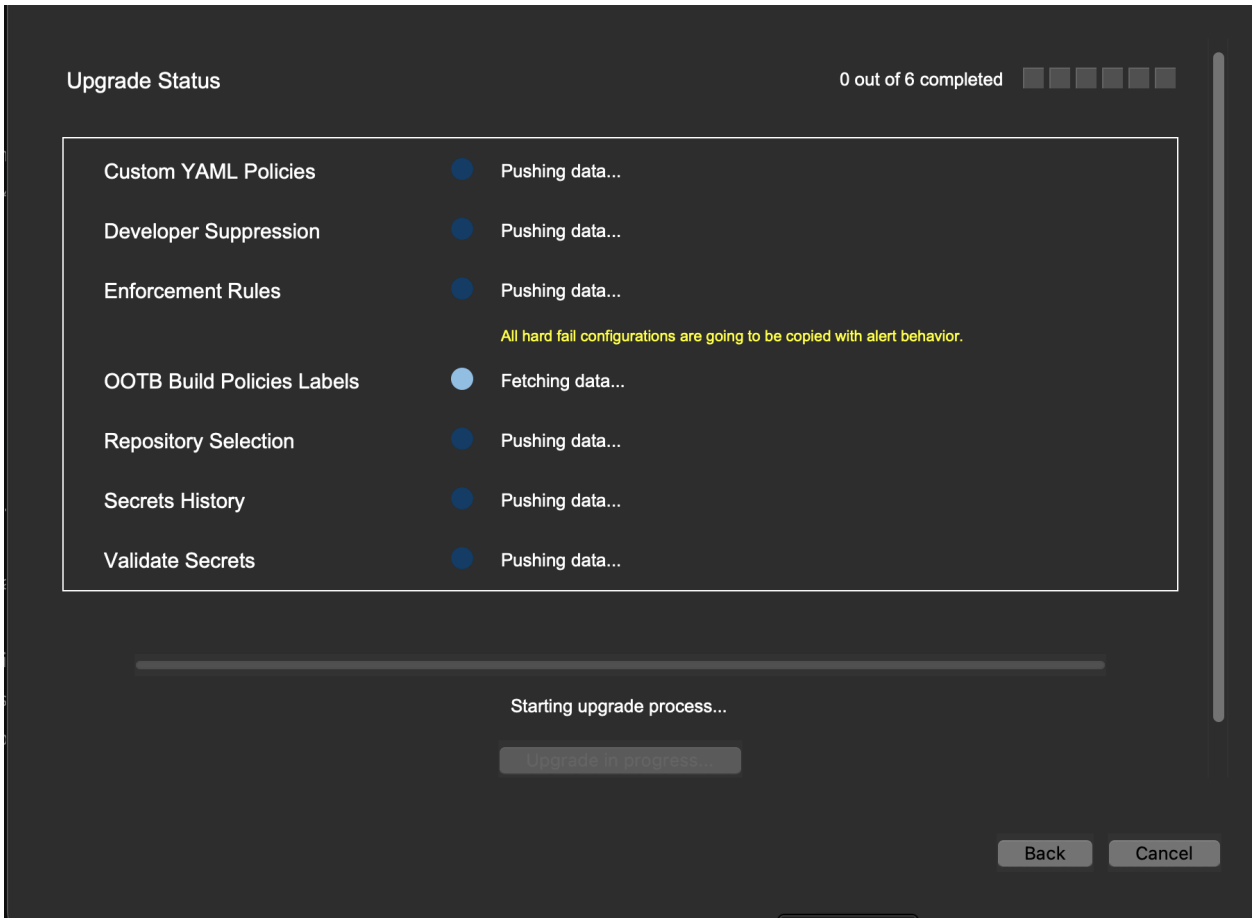
After finalizing your choices in the **Content Selection** section, click Next to proceed to the **Review** tab. This screen provides a summary list of the features you've selected to copy from Prisma Cloud into Cortex Cloud. Consider this your final confirmation step before initiating the actual upgrade process.

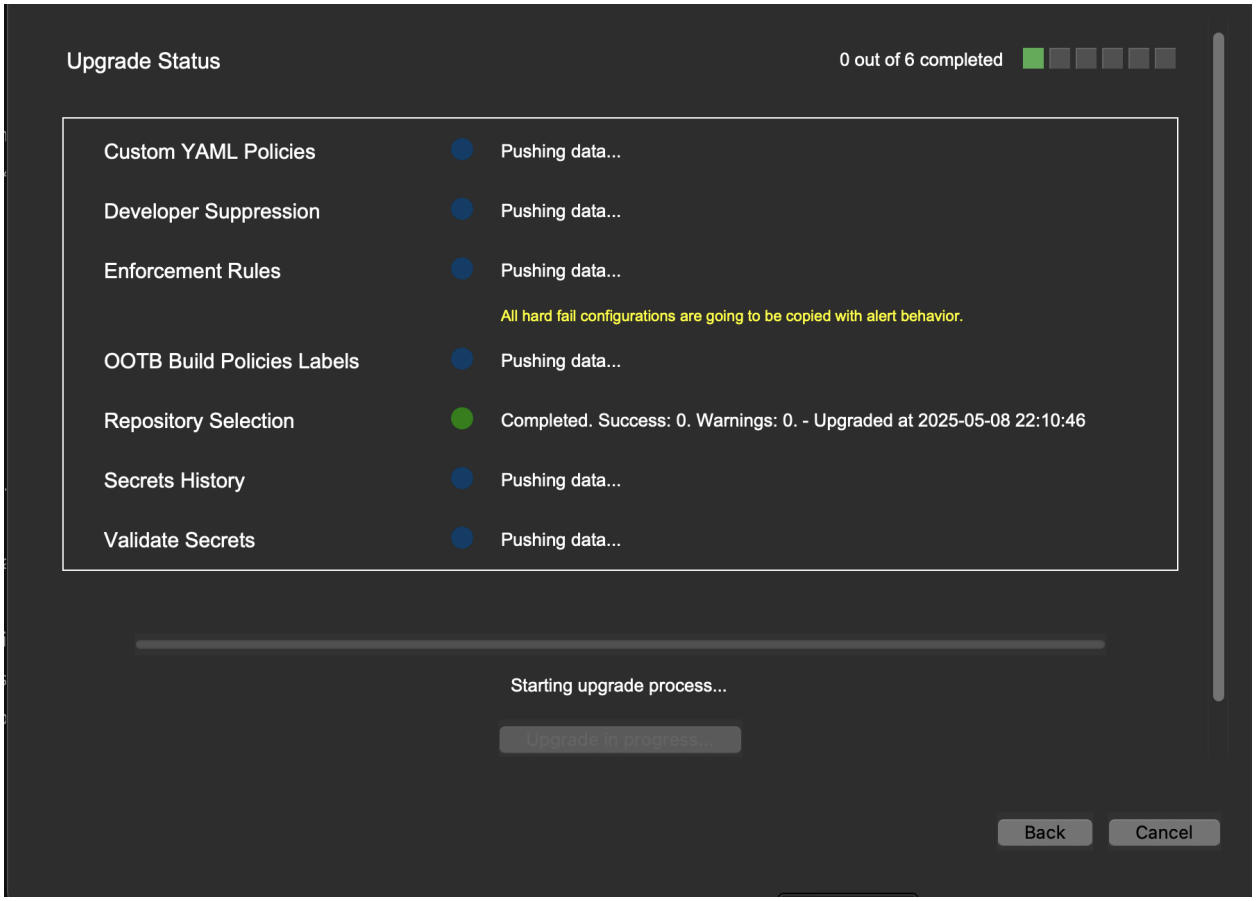
Carefully review the summarized selections to ensure they are exactly what you intend to upgrade.

Important: Once the upgrade process begins, it is strongly advised not to interrupt it. Doing so can lead to incomplete transfers or discrepancies in your Cortex Cloud environment. Verify everything is correct before starting the upgrade.



Once the upgrade begins, the **Review** screen dynamically updates to show the real-time progress for each feature you selected. You can observe the status of individual items as they move through the upgrade workflow.





Please be aware that during the execution process, labels may be updated, included, or removed from the view. This is expected behavior. If you observe a discrepancy between the number of resources processed during the transformation and the final execution results, it is likely due to restrictions, considerations, or the scope of the tool, which may prevent all processed elements from being pushed into Cortex.

If the tool appears frozen for an extended period (e.g., 2+ hours) without progress and the logs are not showing any recent update, or if it closes before completing its tasks, please document the issue and reach out via a TAC case for assistance.

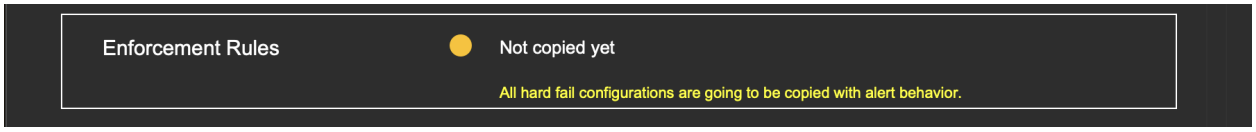
Upgrade Stages and Status Monitoring

The tool processes each feature through several key stages, generally involving: Ingestion (reading from Prisma Cloud), Transformation (adapting data for Cortex

Cloud), and Push (writing data to Cortex Cloud). The status displayed next to each feature indicates its current progress:

- **Not copied yet:** The upgrade for this feature is queued and has not yet started.
- **Fetching Data:** Actively retrieving the required information from Prisma Cloud.
- **Transformation Complete:** Data has been successfully fetched and prepared in the format needed for Cortex Cloud.
- **Pushing data:** Currently transferring the prepared data into your Cortex Cloud environment. This contains a progress bar to show what's the percentage of progress the logs will contain the detail of the results of each information sent into Cortex Cloud.
- **Complete:** The feature was successfully migrated and written to Cortex Cloud.
- **Error:** An issue occurred while processing this feature. Check logs or tool messages for more details.

Important: Be aware that, by design, any policies or rules from Prisma Cloud configured to perform prevention or blocking actions (e.g., Enforcement Rules set to 'Block' or 'Hard Fail') will be migrated to Cortex Cloud in an “Alert-only” mode or “Disabled”. This ensures that automated blocking does not immediately take effect in your new environment, allowing you time for review and deliberate activation within Cortex Cloud post-upgrade. As shown in the example below, an Enforcement Rule set to 'Hard Fail' will be copied in alert behavior.



Once the upgrade process for all selected features is complete, the **Review** tab will display the final **Upgrade Summary**. This summary provides a high-level overview of the entire operation's outcome, typically indicating the overall status (e.g., 'Completed Successfully', 'Completed with Errors', or ‘Error’) and often includes counts of features successfully copied versus those that encountered errors. It's recommended to review this summary and consult the detailed logs (if errors occurred) before proceeding to verify the results directly within your Cortex Cloud

environment. Review the Logs section of this User Guide on how to analyze the same.



Note that multiple executions of the tool are going to update the elements created previously as long as their name or identity remains the same.

Troubleshooting

This section assists in diagnosing and resolving issues encountered while using the Self-Service Upgrade Tool, which utilizes the Cortex Cloud APIs and the Prisma Cloud APIs. Common challenges during upgrade can arise when making API calls, interpreting server responses (including specific HTTP status codes and error messages returned by the server), handling data transformations prior to sending, and ensuring successful data ingestion into Cortex Cloud. Below, we outline frequent error messages and scenarios related to these areas, offering guidance and potential solutions. For complete endpoint specifications, request/response formats, and authoritative error code details, always refer to the official Cortex Cloud API documentation linked above.

Main Considerations During Troubleshooting

- **Authentication and Authorization:** Verify that the API credentials (e.g., API keys, tokens) are valid, correctly formatted in the request headers or parameters, and have the necessary permissions for the intended operation. Look for **401 Unauthorized** or **403 Forbidden** status codes.
- **Request Payload and Formatting:** Ensure that the request body is correctly formatted (e.g., valid JSON), includes all mandatory fields, and adheres to the data types and structures defined in the API specification. **400 Bad Request** errors frequently point to payload issues.
- **Server Responses and Error Codes:** Don't just look at the success/failure of the call; examine the HTTP status code and the response body. The body often contains specific error codes or messages from Cortex Cloud detailing the exact problem (e.g., invalid field value, resource conflict **409 Conflict**, server-side issues **5xx Server Error**).
- **Data Transformation Logic:** If your process involves transforming data before sending it to the API, meticulously review this logic. Ensure the output precisely matches the schema expected by the Cortex Cloud API endpoint. Subtle errors here can lead to **400 Bad Requests** or unexpected behavior during data ingestion.
- **Data Ingestion Confirmation:** Even if an API call returns a success status (e.g., **200 OK, 202 Accepted, 204 No Content**), verify that the data has been correctly ingested and processed within Cortex Cloud as expected. Check relevant logs or the Cortex Cloud interface.
- **Rate Limiting:** Be mindful of API rate limits. Exceeding allowed request quotas can result in **429 Too Many Requests** errors.

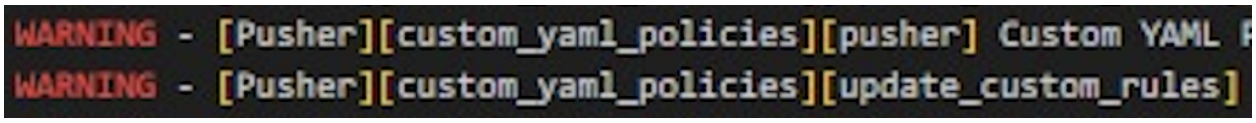
- **Network Connectivity:** Rule out basic network connectivity issues between the client making the call and the Palo Alto Networks Cortex Cloud API endpoints and Prisma Cloud APIs.
- **Data already exists:** There are some situations where features selected are already present in the Cortex Cloud Console, the reason why the Upgrade tool is not going to be able to do the copy process, and it will log them with the corresponding case (e.g., **Skipping creation - policy already exists with that name**).

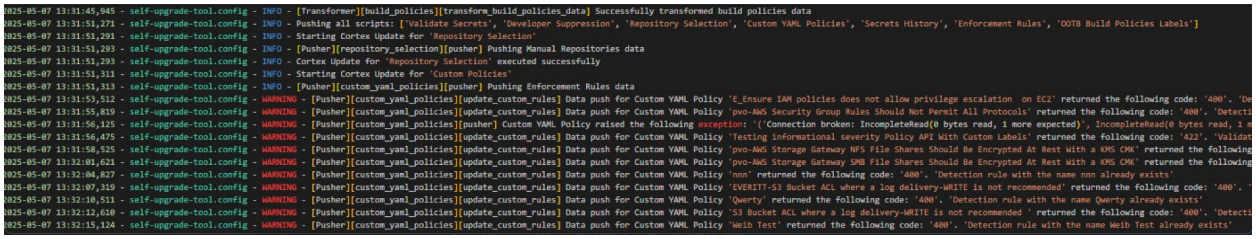
Logs

The Self-Service Upgrade Tool generates a dedicated log file named **self-upgrade-tool.log** during execution. This essential log, found within the primary logs folder "***<log directory tied to the user>/self-upgrade-tool/***", contains a detailed record of all steps and actions performed during the upgrade process. Consult this log file to track the execution sequence, or to assist in diagnosing any errors or unexpected behavior encountered, for example:

- On Windows: "C:\Users\<User Home Directory>\AppData\Local\Logs"
- On Mac: "/Users/<User Home Directory>/Library/Logs/self-upgrade-tool"

When analyzing this log file, pay attention to the structure of the entries. Information is often categorized by the specific tool component or process stage, and the feature or data type being handled. For example, to find information related to the transformation of **Custom YAML Policies**, look for log lines containing identifiers similar to [**Pusher][custom_yaml_policies][pusher]**. This structured logging helps you quickly isolate messages relevant to a specific area or problem.





Outputs

Upon execution, the Self-Service Upgrade Tool produces several outputs designed to clarify the data migration process into Cortex Cloud, including details on ingestion and transformation. These output files are consolidated within a designated folder created at "***<directory tied to the user>/self-upgrade-tool/***". For example:

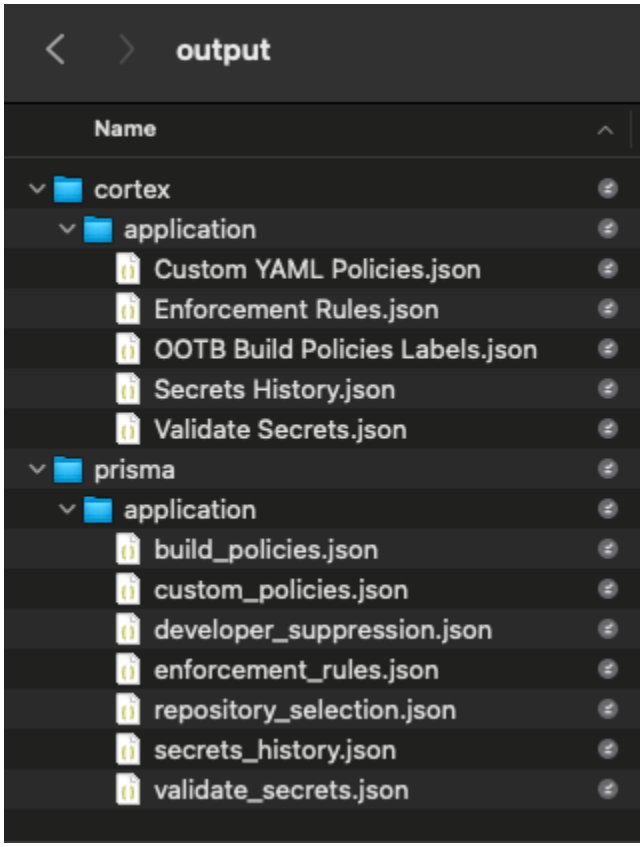
- On Windows: "C:\Users\<User Home Directory>\AppData\Local\self-upgrade-tool"
- On Mac: "/Users/<User Home Directory>/Library/Application Support/self-upgrade-tool"

This folder is structured with two subdirectories: **cortex**, containing information pertinent to the Cortex Cloud environment, and **prisma**, with data related to the source Prisma Cloud environment.



The structure within both the prisma and cortex output folders is organized by the main Prisma categories that can be migrated. Initially, reflecting current tool capabilities, only the **application** subfolder will be present in each directory. JSON files located under **prisma/application** hold the original data ingested from Prisma, while the corresponding files under **cortex/application** contain the data once transformed for Cortex Cloud compatibility.

A comprehensive output in JSON format will also be placed in this folder. The development team will utilize this data and the other available files for troubleshooting and resolving any unforeseen issues.



Document Properties

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Revision History

Date	Revision	Changes By	Status	Comments
04/20/2025	1.0	Andres Rodriguez	In progress	Initial Version of the User Guide for the Self-Service Upgrade Tool
05/08/2025	2.0	Andres Rodriguez	Complete	Final v1.0 release
08/22/2025	3.0	Juan Quinonez	In progress	Initial Version of the User Guide for the July Release of Cortex Cloud.
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12/05/2025	6.0	Andres Rodriguez	Complete	Final v1.3 release